

CUSTOMER CHURN ANALYSIS

Business Analytics Report

AnalystLab Africa Data Analytics Internship Programme

Week 1 Assignment

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Tools Used

Python
Jupyter Notebook
Pandas
NumPy
Matplotlib
Seaborn

Dataset

Telco Customer Churn Dataset

Submission Date

August 6th 2026

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Executive Summary

ABC Communications Ltd. engaged the data analytics team to investigate customer churn and identify factors affecting customer retention. Customer churn reduces recurring revenue and increases the cost of acquiring new customers, making it an important business challenge.

This report presents an analysis of the Telco Customer Churn dataset using Python and Jupyter Notebook. The analysis included dataset inspection, data cleaning, exploratory data analysis, and business visualization. The objective was to identify customer segments with higher churn rates and recommend strategies to improve retention.

The findings indicate that customers on month-to-month contracts, customers with shorter tenure, and customers paying higher monthly charges are more likely to leave the company. Service-related factors, particularly technical support, also influence churn behaviour. Based on these findings, several recommendations are proposed to improve customer loyalty and reduce churn.

Business Understanding

Business Understanding

Customer churn occurs when customers stop using a company's products or services. High churn rates negatively affect business growth because retaining existing customers is generally less expensive than acquiring new ones.

The objective of this analysis was to understand customer behaviour, identify patterns associated with churn, and provide actionable recommendations that will help ABC Communications Ltd. improve customer retention and profitability.

Business Objectives

- Understand the customer base.
 - Identify customer segments with high churn.
 - Evaluate the impact of contract type on retention.
 - Examine the relationship between customer tenure and loyalty.
 - Identify services associated with churn.
 - Assess the effect of payment methods on churn.
 - Recommend practical retention strategies.
-

Dataset Description

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The analysis used the **Telco Customer Churn** dataset provided as part of the internship assignment.

Dataset Summary

Feature	Value
Original Records	7,043
Cleaned Records	7,032
Variables	21
Duplicate Records	0
Blank Values Removed	11

Variables Included

The dataset contains customer demographic information, subscribed services, billing information, payment methods, and contract type, monthly charges, total charges, customer tenure, and churn status.

Dataset Inspection

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	str
1	gender	7043 non-null	str
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	str
4	Dependents	7043 non-null	str
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	str
7	MultipleLines	7043 non-null	str
8	InternetService	7043 non-null	str
9	OnlineSecurity	7043 non-null	str
10	OnlineBackup	7043 non-null	str
11	DeviceProtection	7043 non-null	str
12	TechSupport	7043 non-null	str
13	StreamingTV	7043 non-null	str
14	StreamingMovies	7043 non-null	str
15	Contract	7043 non-null	str
16	PaperlessBilling	7043 non-null	str
17	PaymentMethod	7043 non-null	str
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	str
20	Churn	7043 non-null	str

dtypes: float64(1), int64(2), str(18)

...		SeniorCitizen	tenure	MonthlyCharges
	count	7043.000000	7043.000000	7043.000000
	mean	0.162147	32.371149	64.761692
	std	0.368612	24.559481	30.090047
	min	0.000000	0.000000	18.250000
	25%	0.000000	9.000000	35.500000
	50%	0.000000	29.000000	70.350000
	75%	0.000000	55.000000	89.850000
	max	1.000000	72.000000	118.750000

The dataset consists of 7,043 customer records and 21 variables. Initial inspection revealed no duplicate records and no missing values detected by `isnull()`. However, the `TotalCharges` column contained 11 blank entries stored as text, which required cleaning before analysis.

Data Cleaning

Original Dataset

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Checked Data Types

|



Converted Total Charges to Numeric

|



Removed 11 Blank Records

|



Clean Dataset (7,032 Rows)

Exploratory Data Analysis

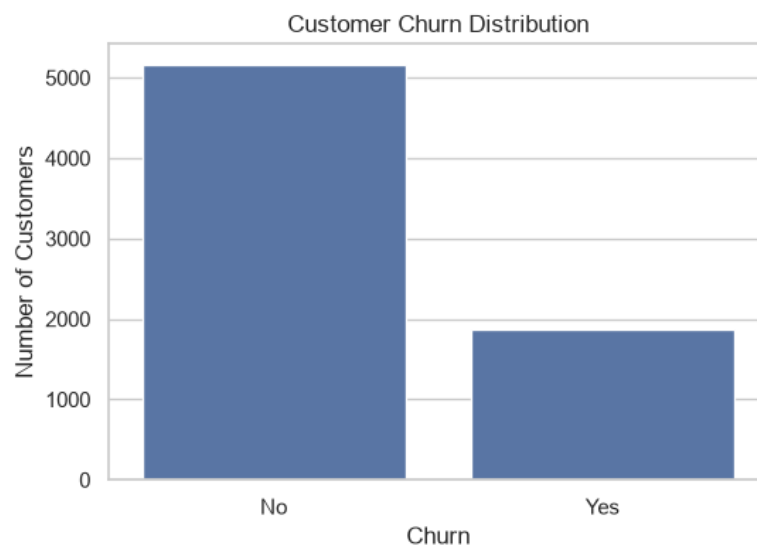
Figure 1. Customer Distribution by Gender



Interpretation

The customer base is almost evenly divided between male and female customers. This indicates that the company serves both genders equally, suggesting that gender alone may not be a major factor influencing customer churn.

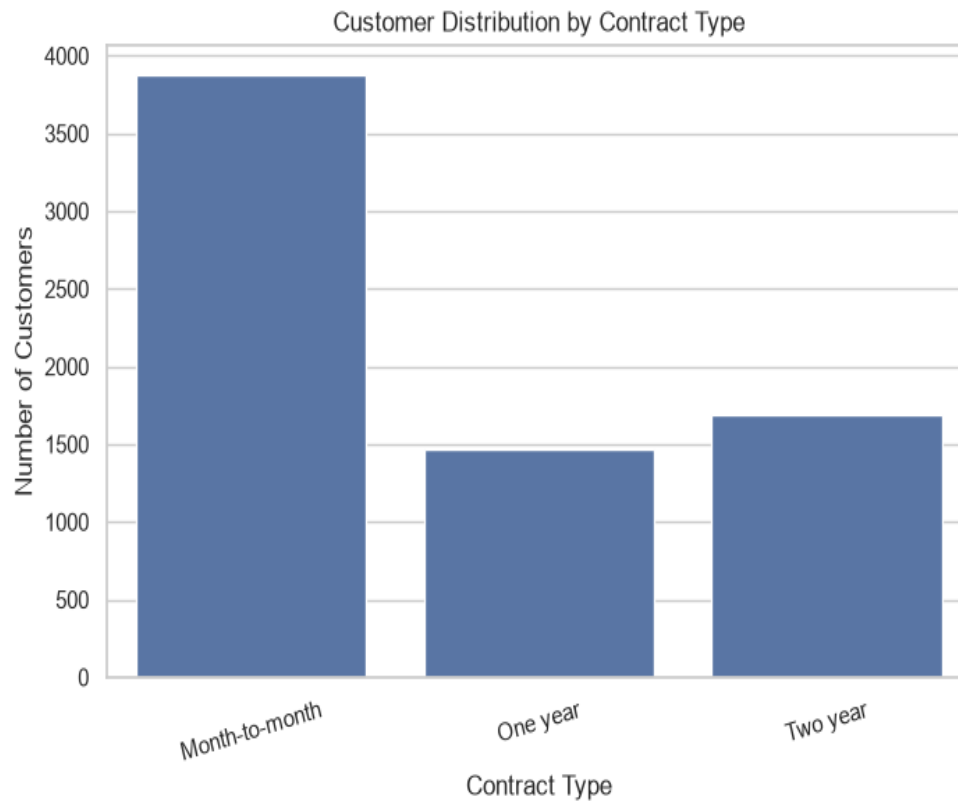
Figure 2. Customer Churn Distribution



Interpretation

The majority of customers did not churn, while a smaller portion left the company. This indicates that customer retention is generally good, but understanding the characteristics of customers who churned is important for improving retention strategies.

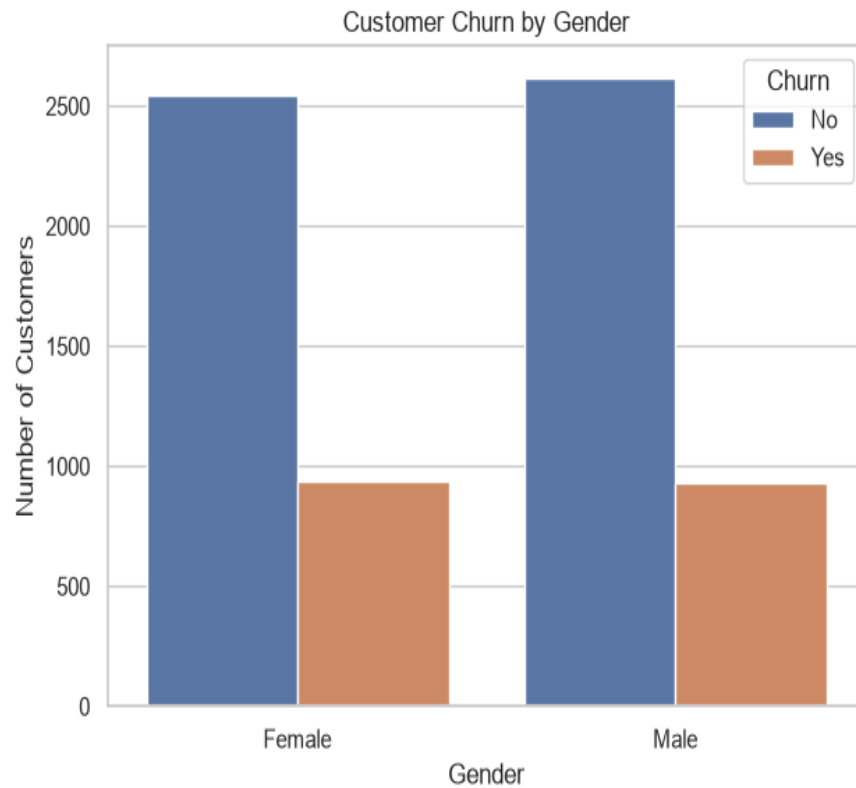
Figure 3. Customer Distribution by Contract Type



Interpretation

This chart shows the popularity of each contract type among customers. The distribution helps identify whether most customers prefer month-to-month, one-year, or two-year contracts.

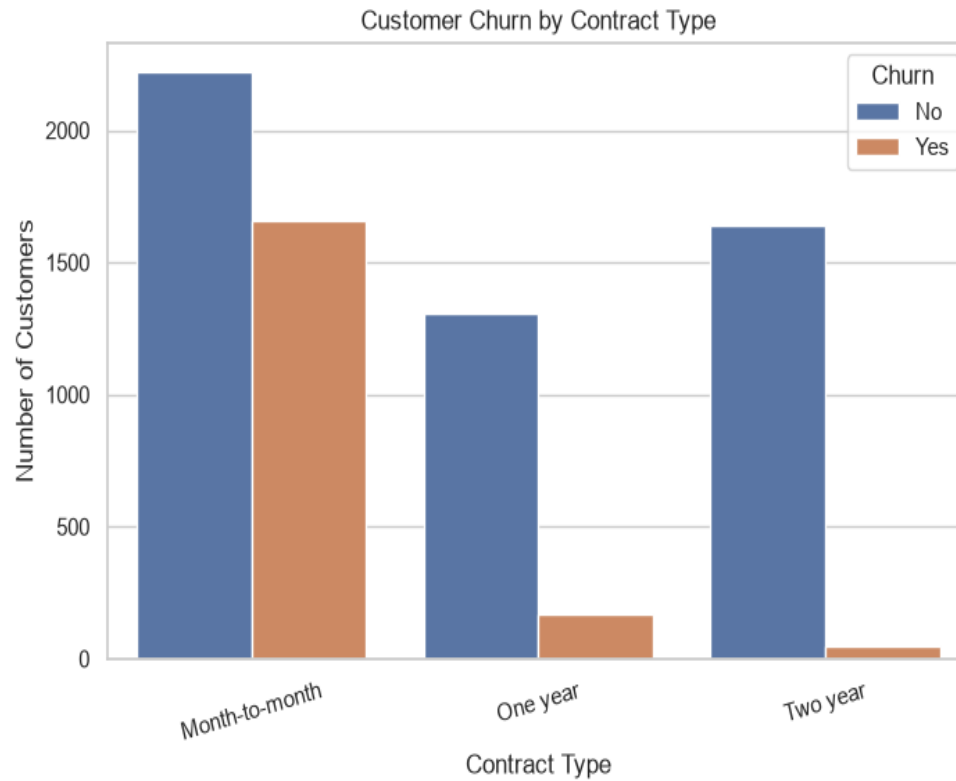
Figure 4. Churn by Gender



Interpretation

This chart compares churn between male and female customers. If both genders show similar churn levels, gender is likely not a major driver of customer churn.

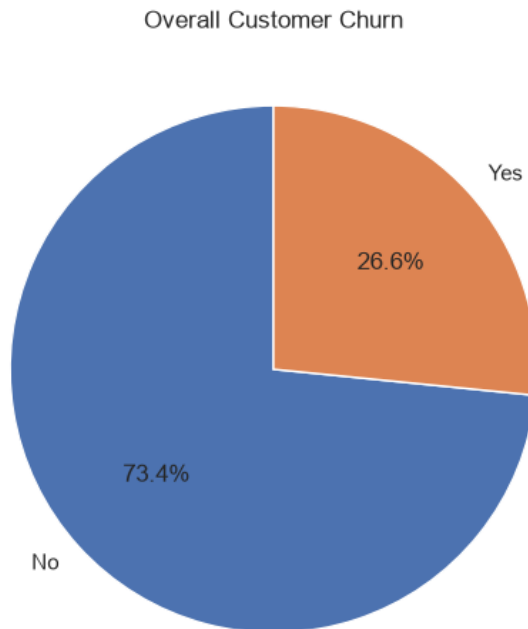
Figure 5. Customer Churn by Contract Type



Interpretation

Customers with month-to-month contracts generally experience higher churn than those with one-year or two-year contracts. Longer contracts appear to improve customer retention.

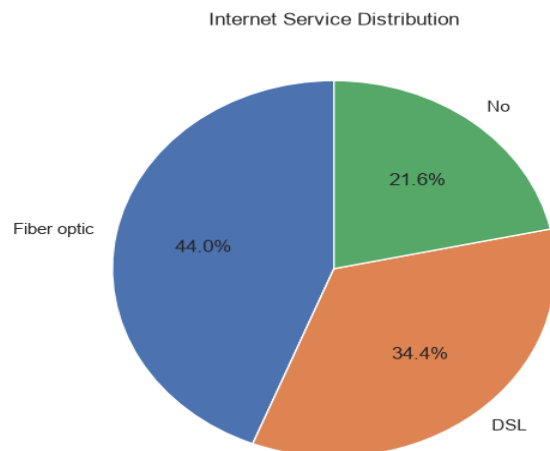
Figure 6. Overall Customer Churn



Interpretation

The majority of customers remained with the company, while a smaller percentage churned. Reducing this churn group can significantly improve long-term business performance.

Figure 7. Internet Service Distribution

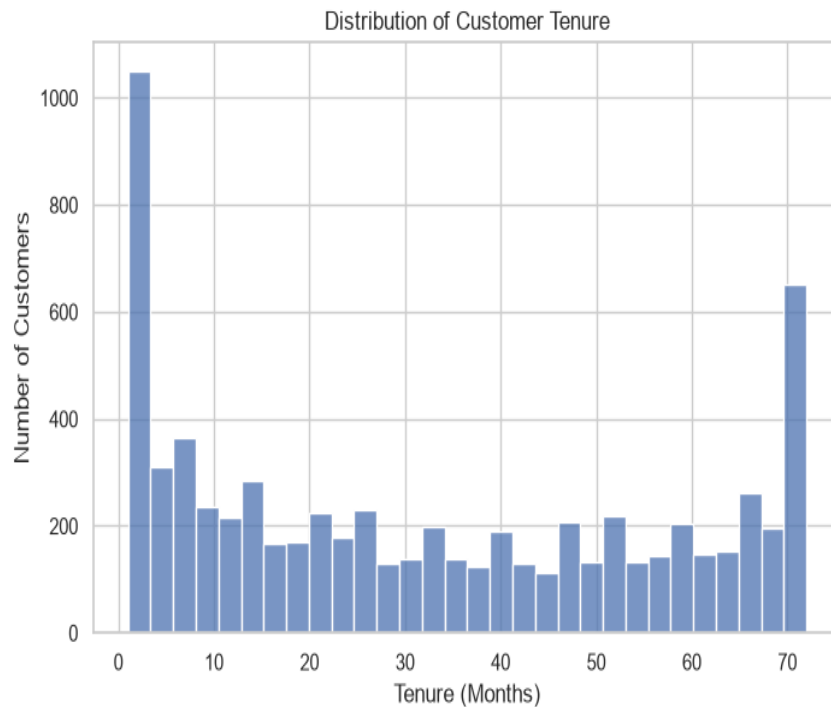


Interpretation

The chart illustrates how customers are distributed across internet service options. This helps identify the company's most commonly used service.

Business Understanding

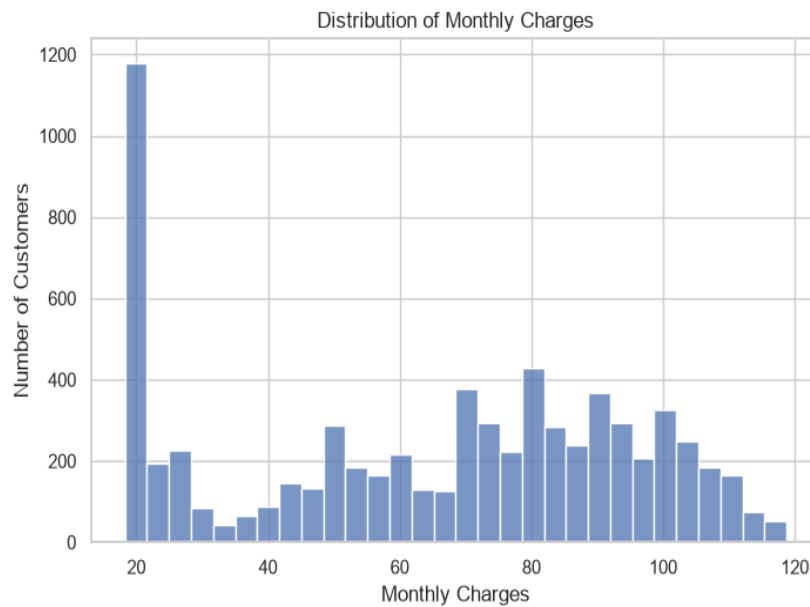
Figure 8. Distribution of Customer Tenure



Interpretation

Most customers have relatively short tenures, while fewer customers remain with the company for many years.

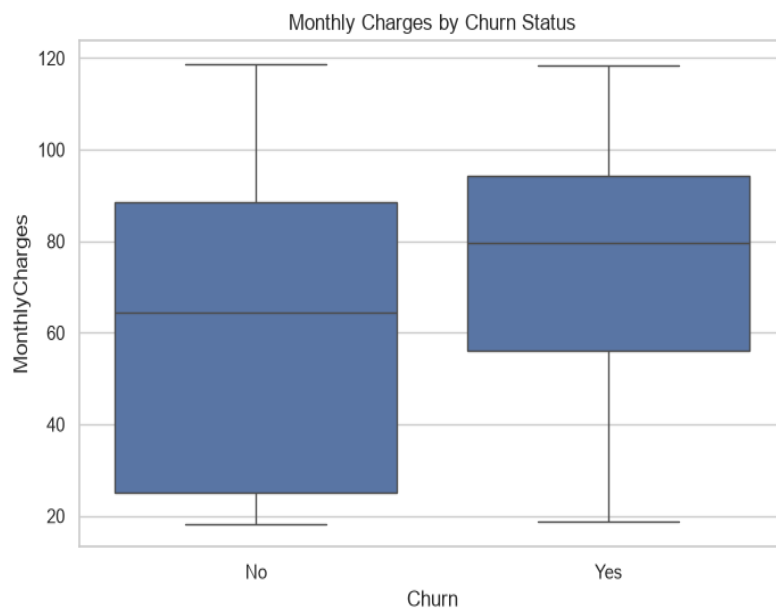
Figure 9. Distribution of Monthly Charge



Interpretation:

Monthly charges vary considerably, with many customers concentrated within the lower and middle price ranges.

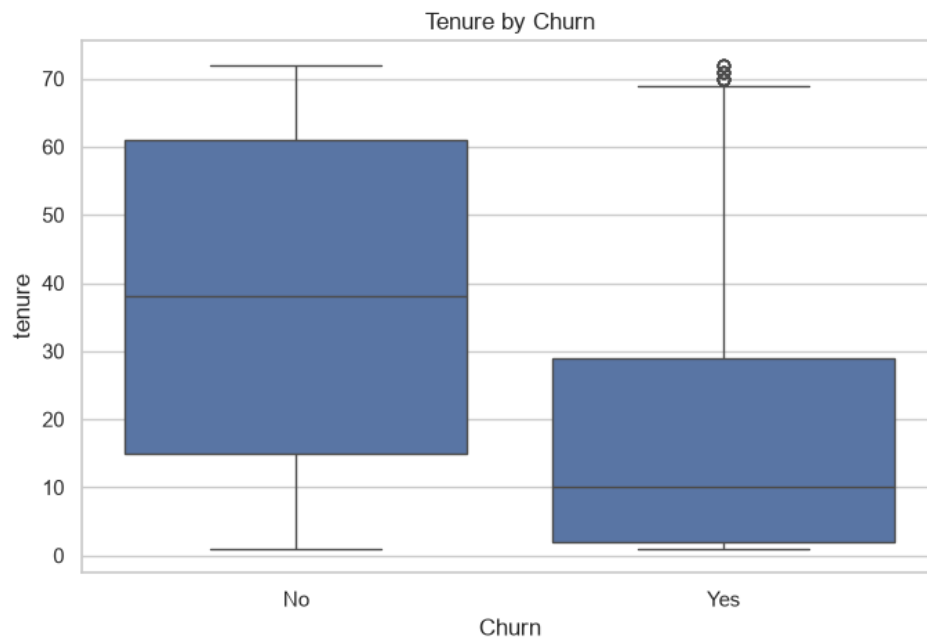
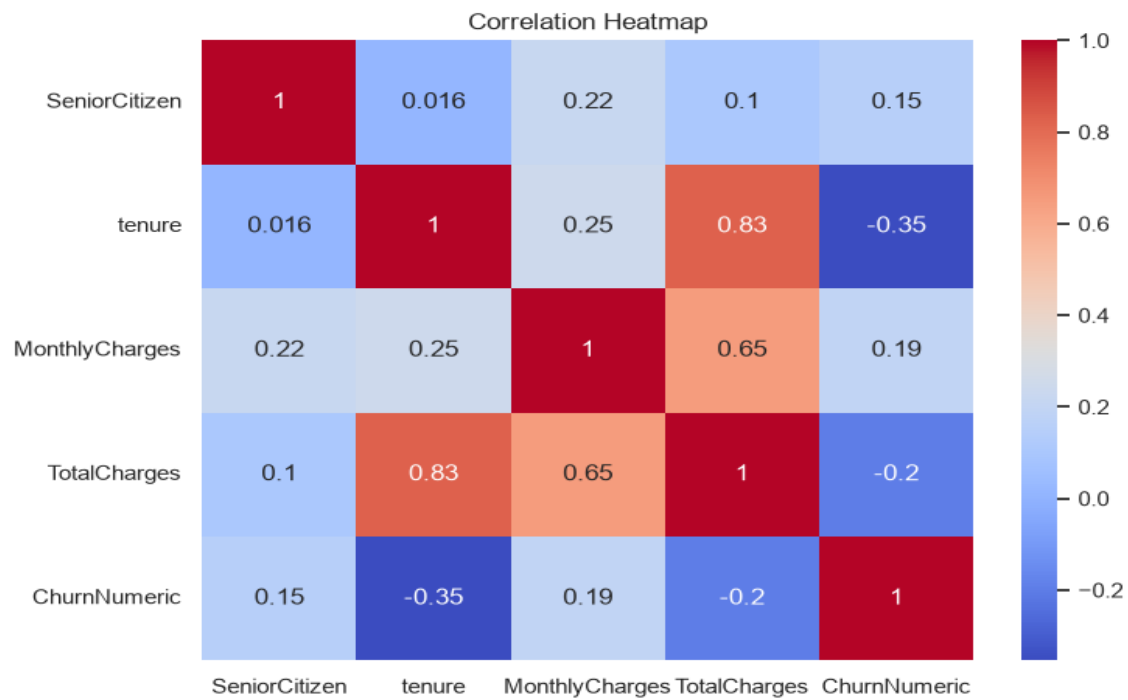
Figure 9. Distribution of Monthly Charges by Churn Status

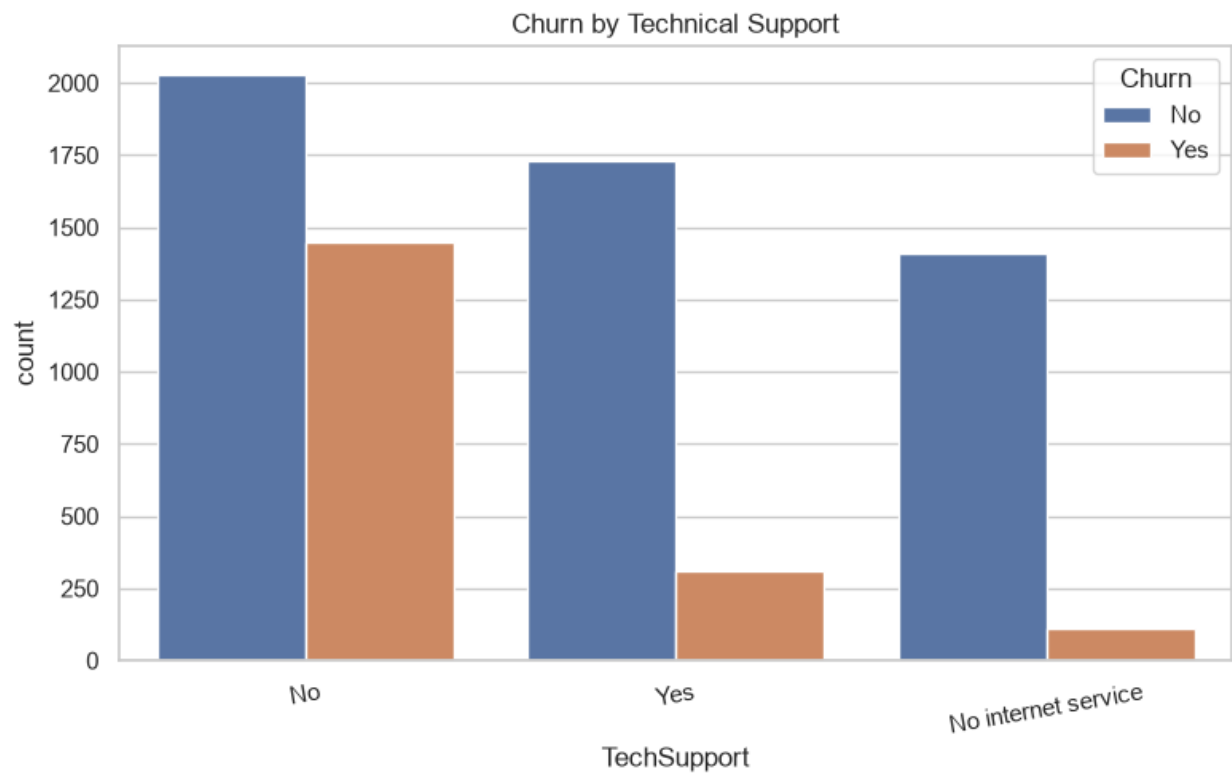
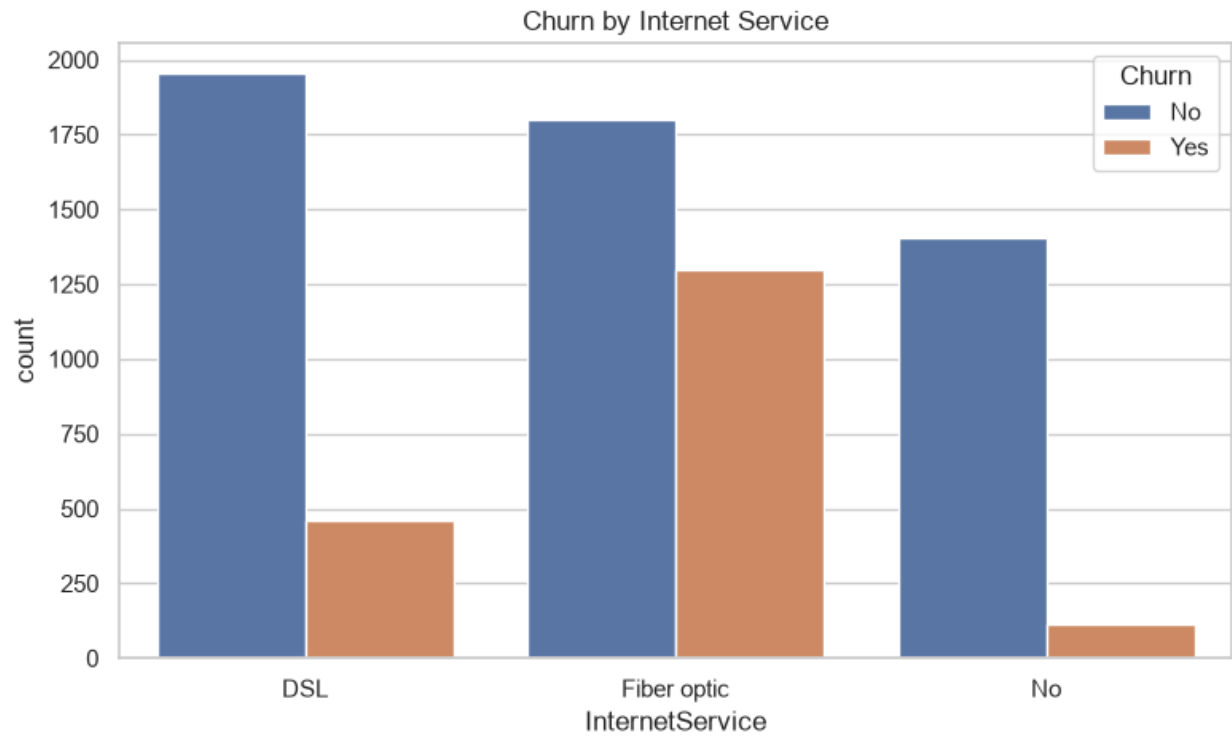


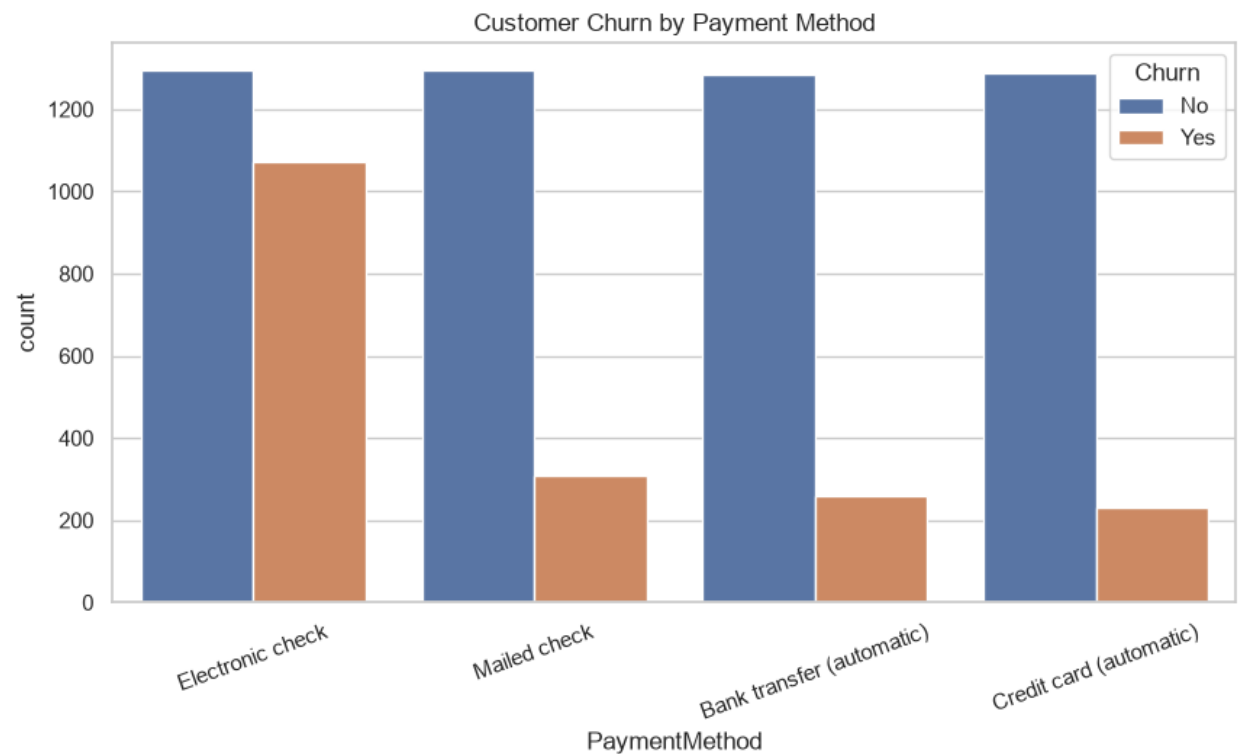
Interpretation:

Customers who churn tend to have higher monthly charges than customers who remain.

Figure 10. Correlation Map







Business Insights

Insight	Business Impact
Customers on month-to-month contracts have the highest churn rate	Flexible contracts provide less incentive for customers to remain with the company.
Customers with shorter tenure are more likely to churn.	The first few months of a customer's relationship with the company are critical for retention.
Customers with higher monthly charges churn more frequently	Pricing may influence customer satisfaction and increase the likelihood of switching providers.
Customers without technical support show higher churn rates	Providing support services appears to improve customer loyalty and satisfaction.
Customers using electronic check payments have relatively higher churn.	Payment preferences may reflect customer engagement and convenience, affecting retention.

Business Risks

Risk	Potential Effect
High churn among month-to-month customers	Reduced recurring revenue and increased customer acquisition costs.
Loss of newer customers	Lower customer lifetime value and slower business growth.
Churn among customers paying higher monthly charges	Loss of higher-value customers and reduced profitability.

Business Opportunities

Opportunity	Expected Benefit
Encourage customers to move to long-term contracts	Increased customer retention and more predictable revenue.
Improve onboarding and customer engagement during the first few months	Better customer satisfaction and reduced early churn.
Promote technical support and value-added services	Higher customer loyalty and stronger long-term relationships.

Business Recommendation

Business Recommendations

1. **Promote long-term contracts** by offering discounts or incentives for customers who switch from month-to-month plans.
2. **Strengthen customer onboarding** with follow-up communications and support during the first six months of service.
3. **Bundle technical support and online security services** with internet plans to improve customer satisfaction.
4. **Review pricing strategies** for customers with higher monthly charges and consider loyalty discounts where appropriate.
5. **Encourage automatic payment methods** by offering small incentives, making payments more convenient and improving customer retention.

Conclusion

This project analyzed the Telco Customer Churn dataset to identify factors associated with customer retention and churn. After inspecting and cleaning the dataset, exploratory data analysis was performed to answer the business questions provided in the assignment.

The analysis showed that contract type, customer tenure, monthly charges, technical support, and payment methods are important factors influencing customer churn. Customers on month-to-month contracts, customers with shorter tenure, and those paying higher monthly charges were more likely to leave the company.

Based on these findings, several actionable recommendations were proposed to improve customer retention, increase customer satisfaction, and support long-term business growth. The insights from this analysis can help ABC Communications Ltd. make data-driven decisions to reduce churn and strengthen customer loyalty.

References

- AnalystLab Africa. *Week 1: Business Analytics Case Study – Customer Behaviour & Business Performance Analysis.*
- Kaggle. *Telco Customer Churn Dataset.*
- Python Software Foundation. *Python Programming Language.*
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- Matplotlib Documentation.
- Seaborn Documentation.